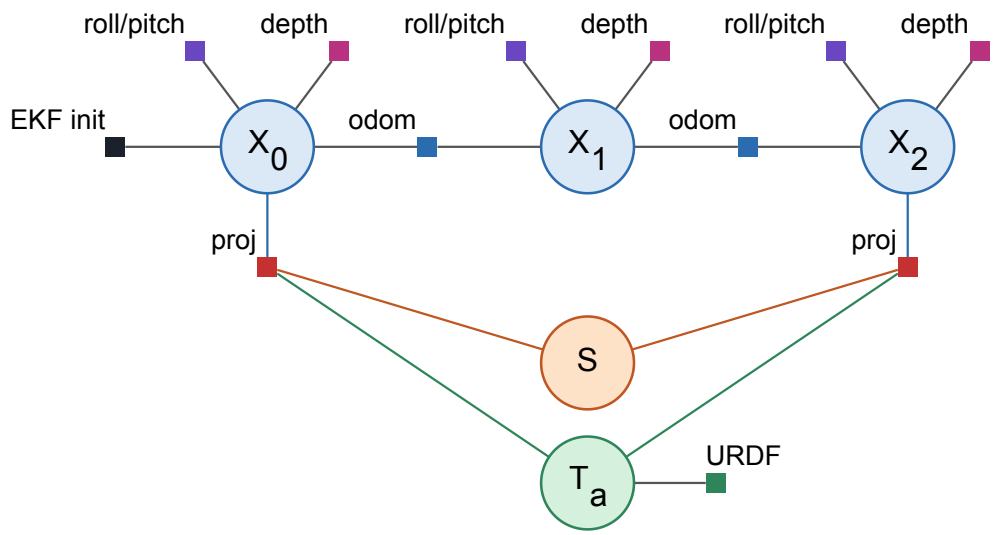
Dock Fuser Factor Graph



One tag, three keyframes: X_0 and X_2 see tag a, X_1 sees none. Circles are variables (solved for); squares are factors (constraints on the variables they connect).

Symbol	Kind	Connects	Meaning
X_k (blue)	variable		AUV pose in odom at keyframe k. Tags pull it off the EKF when the EKF drifts.
S (orange)	variable		Station dock_point pose in odom; one for the whole run. Docking uses $X_k^{-1}S$.
T_a (green)	variable		Tag a's pose relative to S. Stays at its URDF value with one tag.
EKF init (black)	factor	X_0	$X_0 = E_0$, added once. Ties the graph to odom.
odom (blue)	factor	X_{k-1}, X_k	EKF motion $E_{k-1}^{-1}E_k$; relative only, as x, y, yaw drift. Huber.
roll/pitch, **depth**	factor	X_k	Roll/pitch (gravity, yaw free) and depth (pressure) from the EKF.
proj (red)	factor	X_k, S, T_a	4 tag corner pixels vs. prediction of $S \cdot T_a$ seen from $X_k \cdot T_{bc}$. 1 px, Huber.
URDF (green)	factor	T_a	T_a near its station-URDF pose; σ = mounting tolerance.

S and T_a link only through proj, as $S \cdot T_a$ (the tag's pose in odom). E_k: EKF pose at keyframe k. T_{bc}: camera extrinsic. See DOCK_FUSER_PIPELINE.md; figure source dock_fuser_factor_graph.dot.